JanGyan — Learn Without Limits

Category 6: Environment & Civic Responsibility

Module 2 - Waste Management & Cleanliness

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Waste Management & Cleanliness** — how waste is generated, why segregation matters, and the practical systems (composting, recycling, e-waste handling) every household and community can use. Whether you are a student, a homemaker, a working professional, or a civic volunteer, this handbook is written so you can read it once and walk away confident and capable.

By the end of this module, you will not need to search anywhere else to understand the fundamentals of waste management in India.

Learning Objectives

After completing this module, you will be able to:

- Distinguish between the major categories of waste and their sources
- Correctly segregate wet, dry, and hazardous waste at source
- Explain the 3R hierarchy — Reduce, Reuse, Recycle
- Set up basic home composting for organic waste
- Handle e-waste and plastic waste responsibly
- Understand the goals and structure of the Swachh Bharat Mission
- Recognise the government bodies and laws governing waste management in India
- Know exactly how and where to report improper waste disposal

Why This Topic Matters

India generates an enormous volume of waste every single day — from household kitchen scraps to discarded electronics — and the way this waste is handled directly affects public health, groundwater quality, air pollution, and the liveability of our cities and villages.

> ***Reality Check:*** *India generates over 1,60,000 tonnes of municipal solid waste every day (CPCB), yet a significant share still goes untreated and ends up in landfills or open dumps. India is also the world's third-largest generator of e-waste, producing more than 1.6 million tonnes a year — most of it still processed by the informal sector without adequate safety measures.*

Waste management is not just a municipal responsibility — it begins at the household level with a simple habit: **segregation at source**. A single mixed bin can undo the work of an entire city's recycling infrastructure, while a well-segregated one enables composting, recycling, and safe disposal downstream.

This is why JanGyan continues its Environment & Civic Responsibility category with this essential, practical module.

Introduction

Waste is any material that is discarded after its primary use — from food scraps and packaging to broken electronics and construction debris. **Waste management** is the systematic process of collecting, segregating, transporting, treating, and disposing of this waste in a way that protects public health and the environment.

In simple words: waste management is what happens after something leaves your hand and before it either becomes a resource again (through composting or recycling) or is safely disposed of. Poor waste management does not make waste disappear — it simply relocates the problem to a landfill, a water body, or the air we breathe.

Cleanliness, at both the household and community level, is the visible outcome of good waste management practised consistently. It is not a one-time clean-up drive but a daily discipline of segregation, responsible disposal, and shared public accountability.

Fundamental Concepts

Before going deeper, let us understand a few foundational ideas that the entire subject of waste management is built upon.

Categories of Waste

Waste is classified based on its source and composition, each requiring a different handling method.

Category	Description	Examples
Household/Municipal	Everyday waste from homes and public spaces	Food scraps, packaging, sweepings
Biomedical	Waste generated by healthcare facilities	Syringes, bandages, expired medicines
Industrial	Waste generated by manufacturing processes	Chemical sludge, metal scraps
E-Waste	Discarded electrical and electronic items	Phones, batteries, computers
Construction & Demolition	Debris from building activity	Concrete, bricks, wood, metal

The 3R Hierarchy

The internationally recognised waste hierarchy prioritises prevention over disposal, in this order: **Reduce** (avoid generating waste in the first place) → **Reuse** (use an item again for its original or a new purpose) → **Recycle** (process discarded material into new products). Disposal is always the last resort.

Complete Detailed Explanation

Part A — Segregation at Source, In Depth

Segregation at source means separating waste into distinct streams at the point where it is generated — your kitchen, your office desk, your shop counter — rather than mixing everything into one bin. India's Solid Waste Management Rules, 2016 mandate three categories at minimum: wet (biodegradable), dry (recyclable), and domestic hazardous waste.

Bin Colour (Common Convention)	Waste Type	Examples
Green	Wet / Biodegradable	Food scraps, peels, garden waste
Blue	Dry / Recyclable	Paper, plastic, glass, metal
Black / Red	Domestic Hazardous	Batteries, expired medicines, sanitary waste, e-waste

Part B — The Ten Building Blocks of Module 2

This is the heart of the module. Each topic below is explained with its core meaning and its practical relevance for Indian households and communities.

1. Types of Waste

Waste is generally classified by source and composition — household, industrial, agricultural, biomedical, construction, and electronic — each requiring a different collection and treatment pathway under Indian waste management rules.

2. Wet Waste vs Dry Waste

Wet (biodegradable) waste includes food scraps, peels, and garden waste that decompose naturally and can be composted. Dry (non-biodegradable) waste includes plastic, paper, glass, and metal that can typically be recycled. Keeping them separate is the single most important habit in household waste management.

3. Waste Segregation

Segregating waste at the source — using separate bins for wet, dry, and hazardous waste — makes downstream composting and recycling dramatically more efficient and reduces the volume that ends up in landfills.

4. Recycling

Recycling converts discarded paper, plastic, glass, and metal into new products, conserving raw materials and energy while reducing pollution. Clean, dry, properly sorted waste recycles far more efficiently than contaminated or mixed waste.

5. Reusing & Reducing Waste

Reducing consumption prevents waste before it is created — the most effective step in the hierarchy. Reusing items (glass jars, cloth bags, refillable containers) extends their life before recycling or disposal becomes necessary.

6. Composting Basics

Composting converts wet waste into nutrient-rich manure through natural decomposition by micro-organisms. A simple home compost bin or pit reduces landfill waste, cuts methane emissions, and produces free fertiliser for kitchen gardens.

7. E-Waste Management

Electronic waste contains both hazardous materials (lead, mercury) and valuable recoverable metals. Under the E-Waste (Management) Rules, 2022, it must be handed over only to authorised collection centres or recyclers — never mixed with regular household trash.

8. Plastic Waste Management

Plastic waste, particularly single-use plastic, degrades slowly and severely impacts soil, water, and marine life. India banned identified single-use plastic items from 1 July 2022 and requires producers to meet Extended Producer Responsibility (EPR) recycling targets.

9. Swachh Bharat Mission

Launched in 2014, the Swachh Bharat Mission (Clean India Mission) is a national campaign to eliminate open defecation and improve solid waste management across urban and rural India, now in its second phase (SBM-Urban 2.0 and SBM-Gramin Phase II) with a focus on garbage-free cities and villages.

10. Responsible Waste Disposal

Responsible disposal means never littering, always using designated bins, segregating before disposal, never burning waste in the open, and handing over hazardous items like batteries and medicines through proper collection channels.

Important Terminologies

Term	Simple Meaning
Segregation at Source	Separating waste into categories at the point it is generated
Biodegradable	Material that breaks down naturally through microbial action
Non-Biodegradable	Material that does not decompose naturally within a reasonable time
Composting	Natural decomposition of organic waste into nutrient-rich manure
Extended Producer Responsibility (EPR)	A rule requiring producers to manage the post-consumer life of their packaging or products
Material Recovery Facility (MRF)	A facility where mixed recyclables are sorted and processed for reuse
Landfill	A designated site for the disposal of waste by burial
Bioremediation	Treating legacy landfill waste using biological processes
E-Waste	Discarded electrical and electronic equipment
Single-Use Plastic	Plastic items designed to be used only once before disposal

Step-by-Step Understanding

The Journey of Household Waste

1. **Generation** — Waste is created in the kitchen, bathroom, and other parts of the home
2. **Segregation** — Waste is separated into wet, dry, and hazardous bins at source
3. **Collection** — Municipal workers or waste collectors pick up segregated waste
4. **Transportation** — Waste is moved to processing facilities or transfer stations
5. **Treatment** — Wet waste is composted or converted to biogas; dry waste is recycled
6. **Disposal** — Only non-recoverable residual waste is sent to a scientifically managed landfill

How to Set Up Basic Home Composting

1. Choose a bin, pit, or earthen pot with drainage and ventilation
2. Add a layer of dry leaves or soil at the base
3. Add kitchen wet waste in layers, avoiding oily or dairy items
4. Turn the mixture every few days to add air and speed decomposition
5. Keep it moist but not waterlogged
6. Harvest dark, crumbly compost after 4–8 weeks for use in plants and gardens

Real-Life Indian Examples

(Widely known public examples, described for educational awareness.)

- **Indore's Waste Management Model:** Consistently ranked India's cleanest city under the Swachh Survekshan rankings, Indore achieved this largely through strict door-to-door segregated waste collection and public participation.
- **Ambikapur's Zero-Waste Model (Chhattisgarh):** A women-led self-help group network built a citywide door-to-door segregation and material recovery system, turning waste management into a livelihood programme.
- **Alappuzha's Decentralised Waste Model (Kerala):** Instead of large centralised landfills, the city adopted household and community-level composting and biogas units, reducing waste transportation and landfill dependency.
- **EPR Plastic Recycling Progress:** Over 20.7 million tonnes of plastic packaging waste have been recycled under India's Extended Producer Responsibility guidelines since they came into force in 2022 (Ministry of Environment, Forest and Climate Change).

Practical Scenarios

Scenario 1: Your household waste collector asks you to hand over a single mixed bag every day.

Politely request separate wet and dry bins, or ask your Resident Welfare Association to coordinate segregated collection with the municipal body — mixing waste after collection defeats the purpose of segregating it at home.

Scenario 2: An old mobile phone and a set of dead batteries have been sitting in a drawer for months.

Do not throw these in regular trash. Locate an authorised e-waste collection point or a manufacturer take-back programme — many electronics retailers now offer drop-off facilities.

Scenario 3: A neighbour regularly burns dry leaves and plastic waste in an open lot.

This is both an environmental violation and a health hazard. Raise it with the person directly if safe to do so, and report it to your local municipal or pollution control authority if it continues.

Government Rules / Guidelines

- **Ministry of Housing and Urban Affairs (MoHUA):** Administers Swachh Bharat Mission (Urban) and its solid waste management components in cities and towns.
- **Department of Drinking Water and Sanitation, Ministry of Jal Shakti:** Administers Swachh Bharat Mission (Gramin) for rural sanitation and solid/liquid waste management.
- **Central Pollution Control Board (CPCB):** Issues annual reports and monitors compliance under the Solid Waste Management, Plastic Waste Management, E-Waste, and Bio-Medical Waste Rules, with State Pollution Control Boards (SPCBs) enforcing at the state level.
- **NITI Aayog:** Projects national waste generation trends and drives the Waste-to-Wealth mission for resource recovery.

Laws & Policies

Law / Policy	What It Covers
Solid Waste Management Rules, 2016	Mandates source segregation into wet, dry, and hazardous categories; applies to every waste generator, from households to institutions
Plastic Waste Management Rules, 2016 (as amended 2021, 2022)	Bans identified single-use plastic items (since 1 July 2022) and sets Extended Producer Responsibility (EPR) targets
E-Waste (Management) Rules, 2022	Requires producers to meet phased e-waste recycling targets and mandates authorised collection/recycling channels
Bio-Medical Waste Management Rules, 2016	Regulates the segregation, treatment, and disposal of waste generated by healthcare facilities
Construction and Demolition Waste Management Rules, 2025	Sets recycling mandates for construction and demolition debris, effective April 2026
Environment (Protection) Act, 1986	The umbrella law empowering government action against any activity, including improper waste disposal, that harms the environment

> *Under the Solid Waste Management Rules, 2016, every waste generator — including individual households — is legally responsible for segregating waste into a minimum of three streams before handing it over for collection.*

Important Statistics

Metric	Figure
India's municipal solid waste generated per day (CPCB)	Over 1,60,000 tonnes
Share of municipal solid waste treated nationally	Approximately 54% (up from 20% in 2014)
Untreated waste going to landfills daily	Over 30,000 tonnes (about 21% of total)
Projected national waste generation by 2030 (NITI Aayog)	Up to 165 million tonnes per year
India's e-waste generation	Over 1.6 million tonnes per year (world's 3rd largest generator)
Share of India's e-waste handled by the informal sector	Around 90%
Plastic packaging waste recycled under EPR since 2022	Over 20.7 million tonnes
E-waste recycling target trajectory (E-Waste Rules, 2022)	60% (2023–25) → 70% (2025–27) → 80% (2027–28 onward)

(Figures sourced from the Central Pollution Control Board (CPCB), NITI Aayog, and Ministry of Environment, Forest and Climate Change (MoEFCC) reports; official portals should be checked for the latest updates.)

Best Practices

- Segregate waste into wet, dry, and hazardous categories every single day
- Compost kitchen waste at home wherever space allows
- Hand over e-waste only to authorised collectors or manufacturer take-back programmes
- Carry reusable bags and containers to avoid single-use plastic
- Rinse and dry recyclables before placing them in the dry-waste bin
- Report illegal dumping or open burning to your municipal authority

Common Mistakes

- Mixing wet and dry waste into a single bag or bin
- Throwing batteries, medicines, or e-waste into regular household trash
- Burning garbage or dry leaves in the open
- Assuming a municipal worker will re-segregate waste that was handed over mixed
- Treating a single clean-up drive as a substitute for daily segregation habits

Myths vs Facts

Myth	Fact
"Segregating waste at home doesn't matter if it gets mixed later"	Segregated waste dramatically improves recycling and composting efficiency; mixing later is a system failure to fix, not a reason to skip segregation
"All plastic can be recycled"	Many low-grade or multi-layered plastics cannot be economically recycled and require alternative disposal
"Composting at home is complicated and smelly"	A well-maintained compost bin with the right wet-to-dry ratio produces little odour and is manageable in small spaces
"E-waste is not that different from regular electronics trash"	E-waste contains hazardous heavy metals and must go through authorised recycling channels, never regular waste streams

Do's and Don'ts

Do	Don't
Segregate wet, dry, and hazardous waste daily	Don't mix all waste into a single bag or bin
Compost organic kitchen waste where possible	Don't burn garbage or dry leaves in the open
Hand over e-waste to authorised collectors	Don't throw batteries or e-waste in regular trash
Carry reusable bags and containers	Don't rely on single-use plastic out of convenience
Report illegal dumping to local authorities	Don't assume someone else will report it

Practical Tips

For Households

Keep at least two labelled bins (wet and dry) in the kitchen • Rinse recyclables before disposal • Start a small compost bin for kitchen scraps

For Communities

Coordinate with your RWA for segregated door-to-door collection • Organise periodic e-waste and hazardous-waste collection drives • Set up a community composting pit where feasible

For Schools & Offices

Provide separate labelled bins in common areas • Run periodic awareness sessions on segregation • Set up a paper and e-waste collection point

Practical Checklist

■	I have separate bins for wet and dry waste at home
■	I know where my nearest authorised e-waste collection point is
■	I compost or plan to compost my kitchen waste
■	I avoid single-use plastic in daily shopping
■	I know which government body to contact to report illegal dumping
■	I rinse and dry recyclables before disposal
■	I have discussed waste segregation habits with family or peers

Quick Revision Notes

- Waste management = collection, segregation, treatment, and disposal of discarded material
- The 3R hierarchy: Reduce → Reuse → Recycle, with disposal as the last resort
- Solid Waste Management Rules, 2016 mandate segregation into wet, dry, and hazardous waste
- E-waste and hazardous items must go through authorised channels, never regular trash
- Swachh Bharat Mission drives national sanitation and solid waste management reforms
- CPCB and SPCBs are India's key waste-management regulatory bodies

Key Takeaways

1. Segregation at source is the single most impactful habit in waste management

2. Wet waste can become a resource (compost); dry waste can become a resource (recycled material) — but only if kept separate

3. E-waste and hazardous waste require special, authorised handling — never mix with household trash

4. India has a strong regulatory framework, but its success depends on citizen participation at the household level

5. Cleanliness is not a one-time event — it is a daily discipline practised consistently

Glossary

- **Biodegradable** — Material that decomposes naturally through microbial action
- **Compost** — Nutrient-rich organic matter produced by decomposing wet waste
- **Dry Waste** — Non-biodegradable, typically recyclable waste (plastic, paper, metal, glass)
- **EPR (Extended Producer Responsibility)** — Producer accountability for the post-consumer life of their products or packaging
- **E-Waste** — Discarded electrical and electronic equipment
- **Landfill** — A designated site for burying waste
- **Material Recovery Facility (MRF)** — A site where recyclables are sorted and processed
- **Segregation at Source** — Separating waste into categories at the point of generation
- **Single-Use Plastic** — Plastic items intended for one-time use before disposal
- **Wet Waste** — Biodegradable organic waste such as food scraps and garden trimmings

Frequently Asked Questions

1. What is the difference between wet and dry waste?
Wet waste is biodegradable organic matter (food scraps, peels) that can be composted; dry waste is typically recyclable material (plastic, paper, metal, glass) that does not decompose easily.

2. Why does segregation at source matter so much?
Segregated waste can be efficiently composted or recycled; once different waste types are mixed, separating them again is far more difficult, costly, and often impossible at scale.

3. How do I dispose of old batteries or a broken phone?
Never place them in regular trash. Take them to an authorised e-waste collection point or a manufacturer/retailer take-back programme.

4. Is it safe to compost at home?
Yes. A well-maintained compost bin with a balanced mix of wet waste and dry carbon material (dry leaves, paper) produces minimal odour and is manageable even in small urban spaces.

5. What items are banned under India's single-use plastic rules?
Identified low-utility, high-littering items such as plastic cutlery, straws, certain packaging films, and thin carry bags have been banned since 1 July 2022, under the Plastic Waste Management Amendment Rules.

6. What is Extended Producer Responsibility (EPR)?
A rule requiring producers, importers, and brand owners to take responsibility for collecting and recycling the packaging or products they introduce into the market.

7. What is the Swachh Bharat Mission?
A national campaign launched in 2014 to eliminate open defecation and improve solid and liquid waste management across urban and rural India, now in its second phase.

8. Which law requires households to segregate waste?
The Solid Waste Management Rules, 2016, which apply to every waste generator, including individual households.

9. Who regulates e-waste in India?
The Central Pollution Control Board (CPCB), under the E-Waste (Management) Rules, 2022, which set phased recycling targets for producers.

10. How can I report illegal waste dumping or open burning?
Contact your local municipal corporation, urban local body, or the State Pollution Control Board; many cities also have dedicated civic grievance apps or helplines.


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Official References

- Central Pollution Control Board (CPCB) — annual reports on Solid Waste, Plastic Waste, and E-Waste Management Rules implementation
- Ministry of Housing and Urban Affairs (MoHUA) — Swachh Bharat Mission (Urban) programme information
- Department of Drinking Water and Sanitation, Ministry of Jal Shakti — Swachh Bharat Mission (Gramin) programme information
- NITI Aayog — national waste generation projections and the Waste-to-Wealth mission
- Solid Waste Management Rules, 2016; Plastic Waste Management Rules, 2016 (as amended); E-Waste (Management) Rules, 2022 — India Code portal

Official Government Website Links

- Swachh Bharat Mission (Urban): **https://sbmurban.org**
- Swachh Bharat Mission (Gramin): **https://swachhbharatmission.gov.in**
- Central Pollution Control Board: **https://cpcb.nic.in**
- Ministry of Environment, Forest and Climate Change: **https://moefcc.gov.in**
- Press Information Bureau: **https://pib.gov.in**

Conclusion

Waste management is not a distant policy issue — it happens in every kitchen, every office desk, and every street corner, every single day. The habit of segregating waste at source, composting what can be composted, and disposing of hazardous items responsibly is one of the most direct, measurable ways any citizen can improve public health and environmental outcomes.

Continue your Environment & Civic Responsibility journey with JanGyan. Stay consistent, stay clean, and remember: segregation is where responsible waste management begins.

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Module 2 of the Environment & Civic Responsibility Category